1.1 INTRODUCTION ABOUT ELECTRICITY, METHODS OF POWER GENERATION

This subject is introduced in the higher secondary level. In this subject, students are taught how to maintain and repair electrical appliances and electrical machines and how to connect electrical circuits and repairing minor or major faults in the circuits and motor, rewinding of electrical machines both theoretically and practically. So, by learning this subject, students are able to learn on their own by practicing such learning.

We know that the Earth consists of five big natural element such as water, land, air, fire and sky. The sixth important energy developed by human is called "Electricity". In this modern world, in our day today life, electricity plays a vital role. Simply to say, a man can live without food but cannot live without electricity. Thus in our life, the electrical goods play an important role. Hence the students must know about this source and how it is applied.

All matters whether solid, liquid or gaseous consist of minute particles known as atom. According to modern research, electric current means electrons movement only. So we need to know about atom.

1.1.1 Atom

It has a hard-central core known as nucleus. It contains two types of particles; one is known as proton and carries positive charge. The other is neutron, which is electrically neutral. i.e. it carries no charge. Around the nucleus, in an elliptical orbit, the electrons are revolving. Electrons carry the negative charge. The number of electrons and number of protons in an atom are equal. So the atom is electrically neutral. The number of protons in the nucleus of atom gives the atomic number. The total number of neutron and proton are known as atomic weight, because negligible weight of the electron is not taken to calculate atomic weight.

1.1.2 Methods Of Electricity Production

Electricity is produced by extraction of electrons from an atom. The energies which are used to produce electricity are:

a) Friction
b) Light
c) Pressure
d) Heat
e) Chemical Action
f) Magnetism

a) Electricity due to Friction

Due to the friction of two materials, electrons come out from one material to join with the other material. The material which loses the electron gets (+)ve charge and the material which attracted the electron gets (–)ve charge. This type of electricity is called Static Electricity.

Example :- Materials like glass, rubber, wax, silk, reyon, nylon etc.

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